

Research Interests

My research combines **reinforcement learning** with **psychometric modeling** to simulate human behavior and support individualized ability assessment. I focus on building interpretable, robust RL systems that capture decision-making patterns and enable psychometric analysis in complex environments.

Educational Background

Shanghai University of Finance and Economics	Sep 2022 – Jun 2024
M.S. in Applied Statistics	GPA: 3.58/4 (86.78/100)
China University of Geosciences	Sep 2018 – Jun 2022
B.S. in Statistics	GPA: 4.04/5 (ranking: 1/26)

Awards

The 13th Li Siguang Talent Program	China University of Geosciences	2019
National Scholarship	Ministry of Education of the People's Republic of China	2019
Academician Scholarship	China University of Geosciences	2021
Outstanding Undergraduate Thesis	China University of Geosciences	2022
Outstanding Graduate	China University of Geosciences	2022
Academic scholarship	Shanghai University of Finance and Economics	2022, 2023

Publications

(* authors contributed equally)

- Yu, S.*, **Xu, W.***, et al (2025). Enhancing Prediction Performance through Influence Measure. *The Thirteenth International Conference on Learning Representations (ICLR2025)*.
- Xu, W.**, Peng, R., et al. (2024). Improving Reinforcement Learning via Bayesian Dynamic Ensembles. Submitted.
- Yan W., Feng X., Lin T., et al., including **Xu, W.** (2022). Diverse Subclade Differentiation Attributed to the Ubiquity of *Prochlorococcus* High-Light-Adapted Clade II. *mBio*, 13(1):e03027-21.
- Yang X., Jiang Y., **Xu W.**, et al. (2021). What Motivates People to Share Online Rumors? Deconstructing the Ambiguity of Rumors from a Perspective of Digital Storytelling. *WHICEB 2021 Proceedings*, 66.

Research Experience

InfluenceDriven Training: Enhancing Training Robustness [Webpage]	Mar 2024 – Nov 2024
- Proposed an influence-based metric to quantify the contribution and risk of individual training samples, improving model robustness and generalization. - Developed and evaluated a dynamic active learning and data selection strategy, achieving state-of-the-art performance on multiple benchmark datasets in both simulations and real-world settings. - This work has been accepted to ICLR 2025 (see Publications 1).	
Generalized EM for Psychometric Reinforcement Learning [Webpage]	Nov 2024 – Present
- Designed a Generalized EM framework that integrates reinforcement learning with Item Response Theory models, enabling trait estimation from sequential decision data with large or continuous latent state spaces. - Demonstrated accurate trait recovery and stable value learning on Peg Solitaire tasks, highlighting its potential for scalable and interpretable assessment in educational and psychological applications.	

- Developed a Bayesian Dynamic Ensemble method to improve robustness and convergence efficiency in model-based reinforcement learning (MBRL).
- Achieved notable gains in long-horizon planning and stability across MuJoCo benchmark tasks using the mbrl library.
- Manuscript submitted (see Publications 2).

- Led research exploring Large Language Models as world models in partially observable environments to improve planning and decision-making in reinforcement learning agents.
- Introduced an auxiliary objective to better retain semantic context and applied low-rank adaptation for efficient fine-tuning, which improved generalization and long-horizon performance in RL environments.

Competition Experience

- Built predictive features to analyze product characteristics affecting consumer decisions.
- Designed a decision support system for merchants using forecasting and product attribute optimization.
- Achieved **Top 50/1296** in the preliminary round and **Top 30** in the final round.

- Developed hybrid time-series forecasting models combining gradient-boosted trees and recursive-direct strategies.
- Placed **11th (top 10%)** globally.

Technical Skills & Tools

Programming	Python (NumPy, PyTorch, PyTorch Lightning), R, C++, Bash
Development	Linux, Git, LaTeX, Docker